

CHAPTER ONE: ORDINARY DIFFERENTIAL EQUATIONS

Definition: A differential equation (DE) is an equation that involves one or more derivatives of the dependent variable with respect to one or more independent variables.

Example: a) $\frac{d^2y}{dx^2} + x \frac{dy}{dx} = y - \sin x$ $y = y(x)$

b) $\left(\frac{d^2u}{dt^2}\right)^3 - \frac{d^2u}{dt^2} = u + e^t$, $u = u(t)$

c) $\frac{\partial^2 u}{\partial x^2} = \frac{\partial u}{\partial y} + 2x - y$, $u = u(x, y)$ are differential equations.

Definition: A differential equation that involves only one independent variable is called ordinary differential equation (ODE). Otherwise, we call it partial differential equation (PDE).

Example: In the above example, (a) and (b) are ODE while (c) is PDE

Definition: i) The order of a DE is the order of the highest derivative occurring in the differential equation.

ii) The degree of a differential equation is the highest exponent of the highest order derivative that occurs in the differential equation after the differential equation is expressed as a polynomial of the derivative of the dependent variable.

Example: a) $\frac{dy}{dx} = x^2y + x - 5$ is ODE with order 1 and degree 1

b) $(\ddot{y})^2 = \dot{y} - x^4$ is ODE with order 3 and degree 2

c) $(\dot{y})^{\frac{5}{2}} = \ddot{y} - 1$ is ODE with order 3 and degree 2

d) $k\ddot{y} = (1 + (\dot{y})^2)^{\frac{3}{2}}$ is ODE with order 2 and degree 2

e) $(\dot{y} + x)(y - x\dot{y})^3 = 1$ is ODE with order 1 and degree 4

f) $\frac{\partial^2 u}{\partial x^2} = \frac{\partial u}{\partial y} + 2x - y$ is PDE with order 2 and degree 1

Definition: A relation (function) $y = f(x)$ is a solution of ODE on a given interval I , if it satisfies the ODE, for all x on I .

Example: Let $\dot{y} = -5$ be an ODE then

$y(x) = -5x + 1$ satisfies the given ODE

Hence $y(x) = -5x + 1$ is a solution of the ODE

Also $y(x) = -5x + c$ for any constant c is solution of the ODE

Here $y(x) = -5x + c$ is the general solution. Any solution that can be obtained from it by giving particular values to c is called a **particular solution**.

Initial value problem (IVP) and boundary value problem (BVP)

Most differential equations arise from problems in physics, engineering and science. These equations often have certain conditions only at a point (called **initial conditions**) and at two or more points (called **boundary conditions**) associated with them that they must satisfy. The corresponding problem is known as initial value problem (IVP) and boundary value problem (BVP) respectively.

A solution of IVP or BVP is called particular solution.

Example: Let $\dot{y} = e^x$ be an ODE $y = e^x + c$ is general solution.

a) If the IVP $\dot{y} = e^x, y(0) = 2$ then $c = 1$

Hence $y = e^x + 1$ is a particular solution.

b) If the BVP $\dot{y} = e^x, y(0) = 2, y(1) = e + 1$ then $c = 1$

Hence $y = e^x + 1$ is a particular solution

First order ordinary differential equation (FOODE)

General form is $\dot{y} = f(x, y)$

$$\Rightarrow \frac{dy}{dx} = f(x, y) \Leftrightarrow dy = f(x, y)dx$$

Let $f(x, y) = \frac{-M(x, y)}{N(x, y)} = \frac{dy}{dx} \Leftrightarrow M(x, y)dx + N(x, y)dy = 0$ is the general form of FOODE

Example: a) $\dot{y} = \sin x \cos x$ b) $dy - \ln x dx = 0$ c) $(x-1)y^4 dx - x^3(y^2 - 3)dy = 0$ are FOODEs

CLASSIFICATIONS AND DETERMINATION OF SOLUTION OF FOODES

1. ELEMENTARY FOODE

Form $\dot{y} = f(x)$

$$\Rightarrow \frac{dy}{dx} = f(x) \Leftrightarrow dy = f(x)dx$$

$$\Rightarrow \int dy = \int f(x)dx \Rightarrow y(x) = \int f(x)dx + c \text{ -is the general solution}$$

Example: Solve the DE a) $\dot{y} = \sin x \cos x$

Solution: $\dot{y} = \sin x \cos x \Rightarrow dy = \sin x \cos x dx$

$$\Rightarrow \int dy = \int \sin x \cos x dx$$

$$\Rightarrow y(x) = \int \sin x \cos x dx = \int \frac{\sin 2x dx}{2} = \frac{-\cos 2x}{4} + c - \text{is the general solution}$$

b) Solve the IVP $\dot{y} = x^2 - 1, y(1) = 2$

Solution: $\dot{y} = x^2 - 1 \Rightarrow dy = (x^2 - 1)dx$

$$\Rightarrow \int dy = \int (x^2 - 1)dx$$

$$\Rightarrow y(x) = \int (x^2 - 1)dx = \frac{x^3}{3} - x + c - \text{is the general solution of } \dot{y} = x^2 - 1$$

To find the IVP $y(1) = 2 \Rightarrow c = \frac{8}{3}$

$$\Rightarrow y(x) = \frac{x^3}{3} - x + \frac{8}{3} \text{ is a particular solution of the IVP}$$

Exercise: Solve a) $dy - \ln x dx = 0$ b) $\dot{y} = \sec x$ c) $\dot{y} = e^x, y(0) = 2$

2. SEPARABLE FOODE

General form $\dot{y} = f(x)g(y)$

$$\Rightarrow \frac{dy}{dx} = f(x)g(y) \Leftrightarrow \frac{dy}{g(y)} = f(x)dx$$

$$\Rightarrow \int \frac{dy}{g(y)} = \int f(x)dx \text{ gives general solution}$$

Example: Solve a) $\dot{y} = xy + x - y - 1$

Solution: $\dot{y} = xy + x - y - 1 = x(y+1) - 1(y+1) = (x-1)(y+1) = f(x)g(y)$

Hence the DE is separable

$$\Rightarrow \frac{dy}{y+1} = (x-1)dx \Rightarrow \int \frac{dy}{y+1} = \int (x-1)dx$$

$$\Rightarrow \ln|y+1| + c_1 = \frac{x^2}{2} - x + c_2 \Rightarrow y(x) = -1 + e^{\frac{x^2}{2} - x + c} = ce^{\frac{x^2}{2} - x}$$

Exercise: Solve a) $\dot{y} = e^{x-y}$ b) $\dot{y} = e^{x+y}, y(0) = 0$ c) $dy - (x^2 y^3 - x^2 + y^3 - 1)dx = 0$

d) $(xy^4 - y^4)dx - (x^3 y^2 - 3x^3)dy = 0$

3. HOMOGENOUS FOODE

Definition: A function $f(x, y)$ is said to be homogenous of degree n iff $f(tx, ty) = t^n f(x, y)$, for every real number t .

Example: a) If $f(x, y) = x^2 - xy$ then $f(tx, ty) = t^2 x^2 - t^2 xy = t^2 f(x, y)$

Thus $f(x, y)$ is homogenous function of degree 2

b) if $f(x, y) = g\left(\frac{y}{x}\right)$, then $f(tx, ty) = g\left(\frac{ty}{tx}\right) = g\left(\frac{y}{x}\right) = t^0 f(x, y)$. Thus, $f(x, y)$ is homogenous function of degree 0.

Definition: A differential equation $\dot{y} = f(x, y)$ is homogenous if $f(x, y)$ is homogeneous function of degree 0.

Example: $\dot{y} = \frac{y^2 + xy}{x^2} = \left(\frac{y}{x}\right)^2 + \frac{y}{x} = f\left(\frac{y}{x}\right) = f(x, y)$ is a homogeneous differential equation, since

$f(x, y) = \frac{y^2 + xy}{x^2}$ a homogeneous function of degree 0.

Method of solution

The homogeneous $\dot{y} = \frac{dy}{dx} = f(x, y)$ ----- (*) having the property that $f(tx, ty) = f(x, y)$ can be transformed

into a separable differential equation by making the substitution $v = \frac{y}{x} \Leftrightarrow y = xv, x \neq 0$.

And $\frac{dy}{dx} = v + x \frac{dv}{dx} \Rightarrow \dot{y} = v + x\dot{v}, \dot{v} = \frac{dv}{dx} \Rightarrow v + x\dot{v} = f(v)$

$\Rightarrow \dot{v} = \frac{f(v) - v}{x}, x \neq 0 = \frac{1}{x} g(v)$, where $g(v) = f(v) - v$ which is separable in the variables v and x , and the

resulting equation is solved as a separable differential equation; the required solution of (*) is obtained by back substitution.

Example: Solve a) $\dot{y} = \frac{y^2 + xy}{x^2}$ b) $(x + y)dx - (x - y)dy = 0$

Solution: a) If $f(x, y) = \frac{y^2 + xy}{x^2} \Rightarrow f\left(\frac{y}{x}\right) = \frac{y^2}{x^2} + \frac{y}{x} = \frac{xy^2 + xy}{x^2} = \dot{y}$

Thus, $\dot{y} = f\left(\frac{y}{x}\right)$ is homogenous DE

Hence let $v = \frac{y}{x} \Leftrightarrow y = xv, x \neq 0 \Rightarrow v + x\dot{v} = f(v) = v^2 + v$

$$\Rightarrow \frac{dv}{v^2} = \frac{dx}{x} \Rightarrow \int \frac{dv}{v^2} = \int \frac{dx}{x} \Rightarrow \frac{-1}{v} = \ln|x| + c$$

$$\Rightarrow \frac{-x}{y} = \ln|x + c| \Rightarrow y(x) = \frac{-x}{\ln|x + c|} \text{ is the general solution}$$

Exercise Solve a) $(x + y)dx - (x - y)dy = 0$ b) $x^2 dy = (y^2 - xy)dx$

4. EXACT FOODE

Definition: The DE $M(x, y)dx + N(x, y)dy = 0$ ----- (**) is said to be exact DE iff there exists $U = U(x, y)$

such that $\frac{\partial U}{\partial x} = U_x = M(x, y)$ and $\frac{\partial U}{\partial y} = U_y = N(x, y)$ where $U(x, y)$ is twice continuously differentiable.

$$\begin{aligned} U_x = M &\Rightarrow M_y = U_{xy} \\ \text{Thus } U_y = N &\Rightarrow N_x = U_{yx} \end{aligned} \Rightarrow M_y = N_x \text{ since } U_{xy} = U_{yx} \text{ (} U \text{ is continuously differentiable)}$$

And $U_x dx + U_y dy = Mdx + Ndy = 0 \Rightarrow dU = 0 \Rightarrow U(x, y) = C$ is general solution

Test for exactness: If $M(x, y)$ and $N(x, y)$ are continuous and have continuous first partial derivatives on some region D , then (**) is exact iff $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x} \Rightarrow M_y = N_x$.

Example: consider the differential equation $2xydx + (1 + x^2)dy = 0$. Then $M(x, y) = 2xy$ and $N(x, y) = 1 + x^2$. Since $M_y = 2x = N_x$, the differential equation is exact.

Method of solution

To solve an exact DE

Step 1: Check that $M_y = N_x$ iff the DE is exact. If the DE is exact

$$\text{Step 2: } U(x, y) = \begin{cases} \int Mdx + g(y) \\ \text{or} \\ \int Ndy + g(x) \end{cases} \Rightarrow \begin{cases} U_y = \frac{\partial}{\partial y} \int Mdx + g'(y) = N \\ \text{or} \\ U_x = \frac{\partial}{\partial x} \int Ndy + g'(x) = M \end{cases} \text{ respectively}$$

Step 3: $U(x, y) = C$, where C represents an arbitrary constant, is the general solution.

Example: Solve $y^2 dx + (2xy - 5y)dy = 0$

The DE is not elementary, not separable, and not homogenous

Step1: $M(x, y) = y^2 \Rightarrow M_y = 2y$ and $N(x, y) = 2xy - 5y \Rightarrow N_x = 2y$

Thus $M_y = N_x$ and hence the DE is exact

Step 2: There is $U(x, y)$ such that $U_x = M = y^2 \Rightarrow U(x, y) = \int M dx + g(y) = \int y^2 dx = xy^2 + g(y)$

To find $g(y)$

$$U_y = \frac{\partial}{\partial y}(xy^2) + g'(y) = N = 2xy - 5y = 2xy + g'(y)$$

$$\Rightarrow g(y) = \int -5y dy = -\frac{5y^2}{2} + c$$

$$\text{Thus } U(x, y) = xy^2 - \frac{5y^2}{2} + c$$

Step 3: $U(x, y) = xy^2 - \frac{5y^2}{2} + c = c'$ or $U(x, y) = xy^2 - \frac{5y^2}{2} = C$ is general solution.

Alternatively

Step 2: There is $U(x, y)$ such that $U_y = N = 2xy - 5y \Rightarrow U(x, y) = \int N dy = \int (2xy - 5y) dy = xy^2 - \frac{5y^2}{2} + g(x)$

To find $g(x)$, $U_x = \frac{\partial}{\partial x}(xy^2 - \frac{5y^2}{2}) + g'(x) = M = y^2 = y^2 + g'(x)$

$$\Rightarrow g(x) = \int 0 dx = c, \text{ thus } U(x, y) = xy^2 - \frac{5y^2}{2} + c_1$$

Step 3: $U(x, y) = xy^2 - \frac{5y^2}{2} + c_1 = c_2$ or $U(x, y) = xy^2 - \frac{5y^2}{2} = C$ is general solution.

Exercise: Solve a) $(\sin y + x)dy - (2x - y)dx = 0$ b) $ydx + xdy = 0$ c) $(y - 2x)dx + xdy = 0$

d) $(y^2 - x^3)dx + 2xydy = 0$ e) $2x \sin y + y^2 e^{-x} dx + (x^2 \cos y - 2ye^{-x} + y^2)dy = 0$

5. Non exact DE and integrating factor.

Consider a DE $(2y - 3x)dx + xdy = 0$, then $M = 2y - 3x \Rightarrow M_y = 2$ and $N = x \Rightarrow N_x = 1$

Thus the DE is not exact.

But if we multiply both sides of the equation by x , we get a new DE

$$x(2y - 3x)dx + x^2 dy = 0 \text{ Hence } M = 2xy - 3x^2 \Rightarrow M_y = 2x \text{ and } N = x^2 \Rightarrow N_x = 2x$$

Thus the new DE is exact. Here the function $I(x) = x$ is an integrating factor.

Definition: A nonzero function $\mu(x, y)$ is called an integrating factor (IF) for $M(x, y)dx + N(x, y)dy = 0$ if the DE $\mu(x, y)M(x, y)dx + \mu(x, y)N(x, y)dy = 0$ is exact.

To solve non exact DE

Step 1: Identify $M(x, y)dx + N(x, y)dy = 0$ is not exact

Step 2: Find integrating factor $\mu(x, y)$ if it exists.

Step 3: solve the exact differential equation $\mu Mdx + \mu Ndy = 0$

How to find such $\mu(x, y)$

If $M(x, y)dx + N(x, y)dy = 0$ and $\mu = \mu(x, y)$ is integrating factor, then

$$\mu Mdx + \mu Ndy = 0 \text{ is exact} \Leftrightarrow \frac{\partial}{\partial y}(\mu M) = \frac{\partial}{\partial x}(\mu N)$$

$$\Rightarrow \mu M_y + M\mu_y = \mu N_x + N\mu_x \Leftrightarrow$$

$$M_y - N_x = \frac{1}{\mu}(N\mu_x - M\mu_y)$$

Solving $\mu(x, y)$ is PDE which is not easy. So we consider some special cases:

Case 1: If μ is independent of $y \Leftrightarrow \mu = \mu(x) \Leftrightarrow \mu_y = 0 \Leftrightarrow \frac{M_y - N_x}{N} = \frac{\mu_x}{\mu}$ is independent of y .

$$\int \frac{M_y - N_x}{N} dx = \int \frac{d\mu}{\mu} \Rightarrow \ln|\mu(x)| = \int \frac{M_y - N_x}{N} dx$$

Thus $\mu(x) = e^{\int \frac{M_y - N_x}{N} dx}$ is the IF

Case 2: If μ is independent of $x \Leftrightarrow \mu = \mu(y) \Leftrightarrow \mu_x = 0 \Leftrightarrow \frac{M_y - N_x}{M} = \frac{-\mu_y}{\mu}$ is independent of x .

$$\int -\frac{M_y - N_x}{M} dy = \int \frac{d\mu}{\mu} \Rightarrow \ln|\mu(y)| = -\int \frac{M_y - N_x}{M} dy$$

$\mu(y) = e^{\int \frac{M_y - N_x}{M} dy}$ is the IF

Case 3 and Case 4 are left as exercise.

Example: Solve a) $2xdy + (y-1)dx = 0$

Solution: Step 1: $M = y-1 \Rightarrow M_y = 1$ and $N = 2x \Rightarrow N_x = 2$. Hence the DE is not exact.

Step 2: $\frac{M_y - N_x}{N} = \frac{-1}{2x}$ is independent of y .

Thus $\mu(x) = e^{\int \frac{M_y - N_x}{N} dx} = e^{\int \frac{-1}{2x} dx} = e^{-\frac{1}{2} \ln|x| + C} = ce^{\ln\left(\frac{1}{\sqrt{x}}\right)} = \frac{c}{\sqrt{x}}$ is the IF.

In particular if $C = 1$ $\mu(x) = \frac{1}{\sqrt{x}}$ is an IF

Step 3: The DE $\frac{2}{\sqrt{x}}x dy + \frac{1}{\sqrt{x}}(y-1)dx = 0$ is exact

Hence $N = 2\sqrt{x}$ and $M = \frac{y}{\sqrt{x}} - \frac{1}{\sqrt{x}} \Rightarrow U(x, y) = \int M dx = \int \frac{y-1}{\sqrt{x}} dx = (y-1)2\sqrt{x} + g(y)$

$$\Rightarrow U_y = \frac{\partial}{\partial y} (y-1)2\sqrt{x} + g'(y) = N = 2\sqrt{x} = 2\sqrt{x} + g'(y)$$

$$g(y) = c. \text{ Hence } U(x, y) = (y-1)2\sqrt{x} + c$$

Therefore $U(x, y) = (y-1)2\sqrt{x} + c = C'$ hence $(y-1)2\sqrt{x} = C$ is the general solution.

Exercise: Solve a) $(y^2 - x)dy + (x^2 y^2 + y)dx = 0$: Ans $y + \frac{x}{y} + \frac{x^3}{3} = C$

$$\text{b) } \dot{y} = x + y, y(0) = -3 \quad ; \text{ Ans } e^{-x}y + xe^{-x} + e^{-x} = 1$$

5. LINEAR FOODE (LFOODE)

General form: $\dot{y} + p(x)y = f(x)$ ----- (1)

If the function $f(x) = 0$, then the **LFOODE** is said to be homogenous DE, otherwise, non- homogenous DE. That is, $\dot{y} + p(x)y = f(x)$ ----- (1), for $f(x) \neq 0$ is non- homogenous

$$\dot{y} + p(x)y = 0 \text{ ----- (2), the homogenous part of (1)}$$

From (1), we have $\frac{dy}{dx} + p(x)y = f(x) \Leftrightarrow dy + (p(x)y - f(x))dx = 0$

Here $M = p(x)y - f(x), N = 1 \Rightarrow M_y = p(x), N_x = 0 \Rightarrow M_y \neq N_x$. Thus, the DE (1) is not exact and

$$\frac{M_y - N_x}{N} = p(x) \text{ -is independent of } y. \text{ Hence, the DE (1) has an integrating factor independent of } y$$

$$\Rightarrow \mu(x) = e^{\int \frac{M_y - N_x}{N} dx} = e^{\int p(x) dx} = I(x) \text{ - is the integrating factor.}$$

$\Rightarrow I(x)[\dot{y} + p(x)y = f(x)] \Rightarrow I(x)\dot{y} + I(x)p(x)y = I(x)f(x)$ ----- (3) is exact and can be solved by the method for exact DE.

Multiplying the DE (2) by $I(x) = e^{\int p(x) dx}$ reduces it to the integrable form

$$\frac{dy}{dx} \left(y(x)e^{\int p(x) dx} \right) = q(x)e^{\int p(x) dx} \text{ ----- (4). Integrating (4) both sides with respect to } x, \text{ we get}$$

$$y(x) = e^{-\int p(x)dx} \int e^{\int p(x)dx} f(x)dx + Ce^{-\int p(x)dx}$$

Remarks: 1. $y_1(x) = \frac{1}{I(x)} = e^{-\int p(x)dx}$ -is a fundamental solution of (2).

2. $y_c(x) = cy_1(x) = c_1 e^{-\int p(x)dx}$ -is the general (complementary) solution of (2)

3. $y_p(x) = e^{-\int p(x)dx} \int e^{\int p(x)dx} f(x)dx$ -is the particular solution of (1).

4. $y(x) = y_c(x) + y_p(x) = cy_1(x) + y_p(x)$ -is the general solution of (1)

That is, if y_1 is fundamental solution of (2) and y_p is any solution of (1) then

$y(x) = cy_1(x) + y_p(x) = y_c(x) + y_p(x)$ -is the general solution of (1) (non-homogenous).

Example: Solve 1. $\dot{y} - 3y = 6$

Solution: $\dot{y} - 3y = 6$ ----- (1)

$\dot{y} - 3y = 0$ ----- (2), and $p(x) = -3, f(x) = 6$

$$\Rightarrow \int p(x)dx = -3x \Rightarrow I(x) = e^{\int p(x)dx} = e^{-\int 3x dx} = e^{-3x}$$

$$\Rightarrow y_1(x) = \frac{1}{I(x)} = \frac{1}{e^{-3x}} = e^{3x} \text{ -is the fundamental solution of (2).}$$

$y_c(x) = cy_1(x) = ce^{3x}$ - is the general (complementary) solution of (2). That is, any other solution of (2) is a scalar multiple of y_1 and the particular solution $y_p(x)$ of (1) is given by

$$y_p(x) = e^{-\int p(x)dx} \int e^{\int p(x)dx} f(x)dx = e^{3x} \int 6e^{-3x} dx = \frac{e^{3x} \cdot 6 \cdot e^{-3x}}{-3}$$

$$\Rightarrow y_p(x) = -2$$

Therefore, $y(x) = y_c(x) + y_p(x) = ce^{3x} - 2$ -is the general solution of (2).

$$2. \dot{y} - y = \frac{1}{1 - e^{-x}} = \frac{e^x}{e^x - 1}$$

Solution: The given DE is LFOODE

$$\text{Thus } p(x) = -1, f(x) = \frac{1}{1 - e^{-x}} = \frac{e^x}{e^x - 1}. \text{ Hence } M = -y - \frac{e^x}{e^x - 1}, N = 1$$

$$\Rightarrow \frac{M_y - N_x}{N} = -1 \text{ -is independent of } y.$$

$$\mu(x) = e^{\int -x dx} = e^{-x+c} = Ce^{-x}. \text{ Hence } \mu(x) = e^{-x} \text{ is the integrating factor.}$$

$$\Rightarrow e^{-x}dy + -e^{-x}\left(y + \frac{e^x}{e^x - 1}\right)dx = 0 \text{ -is exact DE}$$

$$\Rightarrow U(x, y) = \int Ndy = \int e^{-x}dy = e^{-x}y + g(x). \text{ Thus } U_x = -e^{-x}y + g'(x) = M = -e^{-x}y - \frac{1}{e^x - 1}$$

$$\Rightarrow g(x) = \int \frac{1}{e^x - 1}dx = \ln\left|\frac{e^x - 1}{e^x}\right| + c \Rightarrow U(x, y) = e^{-x}y + \ln\left|\frac{e^x - 1}{e^x}\right| + c = c'$$

$$\Rightarrow e^{-x}y + \ln\left|\frac{e^x - 1}{e^x}\right| = C - \text{ is the general solution of the DE}$$

Exercise: Solve a) $\dot{y} + \frac{1}{x^2}y = 0$ b) $\dot{y} - 3y = f(x), f(x) = \begin{cases} 1, & x \leq 2 \\ 0, & x > 2 \end{cases}$ c) $\dot{y} + 2xy = 4x$

d) $\dot{y} + 6x^5y = x^5$ e) $\dot{y} - 3y = 15$ f) $x^2\dot{y} - y = 3$ g) $x\dot{y} + y = x \sin x$

Properties of solution of first order linear differential equation

Let $\dot{y} + p(x)y = f(x)$ ----- (1) be the linear first order differential equation, then the related homogenous

DE (RHLFOODE) is $\dot{y} + p(x)y = 0$ ----- (2)

Theorem: If y_1 is a solution of (2), then $y_c(x) = c_1y_1(x)$ is solution of (2) in particular if $c_1 = 0$

$y_c(x) = 0$ - is trivial or singular solution of (2)

Theorem: If y_1 and y_2 are solutions of (2) then $y(x) = c_1y_1(x) + c_2y_2(x)$ is also a solution of (2).

Theorem: If y_1 is non-trivial solution and y_2 is any solution of (2) then $y_2(x) = c_1y_1(x)$ is the general Solution of (2) and $y_c(x) = c_1y_1(x)$ is complementary solution.

Definition: Any non-trivial solution of (2) is called fundamental solution.

Example: Let $\dot{y} - \frac{2}{x}y = 0$, Take $y(x) = x^2$ is fundamental solution

$$\Rightarrow y_c(x) = c_1y_1(x) = c_1x^2 - \text{ is general solution of (2) or complementary solution.}$$

Theorem: If y_1 is fundamental solution of (2) and y_p is any one solution of (1) then

$$y(x) = c_1y_1(x) + y_p(x) = y_c(x) + y_p(x) - \text{ is the general solution of (1) is non-homogenous LFOODE.}$$

7. BERNOLLEOUS DE

Definition: A differential equation that can be written in the form: $\dot{y} + p(x)y = f(x)y^n$ ----- (*),

Where n is real constant, is called a Bernoulli equation.

If $n \neq 0,1$, then a Bernoulli equation is non-linear, but can be reduced to a linear equation as follows. We

first divide equation (*) by y^n to obtain $y^{-n} \frac{dy}{dx} + y^{1-n} p(x) = f(x)$. We now make the change of variables

$u(x) = y^{1-n}$ ----- (**), which implies that $\frac{du}{dx} = (1-n)y^{-n} \frac{dy}{dx}$. That is, $y^{-n} \frac{dy}{dx} = \frac{1}{(1-n)} \frac{du}{dx}$.

Substituting into equation (*) for y^{1-n} and $y^n \frac{dy}{dx}$ yields the linear differential equation

$\frac{1}{(1-n)} \frac{du}{dx} + p(x)u = f(x)$ or in standard form, $\frac{du}{dx} + (1-n)p(x)u = (1-n)f(x)$ ----- (***)

The linear equation (***) can be solved for u as a function of x . The solution to the original equation is then obtained from (**).

Example: Solve a) $\dot{y} + y = xy^2$

Solution: Let $u = y^{1-2} = y^{-1} \Rightarrow \dot{u} = -y^{-2} \dot{y} \Leftrightarrow \frac{\dot{u}}{-1} + u = x$

$\Rightarrow \dot{u} - u = -x$ - is LFOODE and the RHLFOODE is $\Rightarrow \dot{u} - u = 0$

$\frac{du}{u} = dx \Rightarrow \ln|u| = x + c \Rightarrow u_c(x) = ce^x$ and $u_1 = e^x$

Thus $u_p(x) = e^x \int \frac{-x}{e^x} dx = -e^x \int xe^x dx = x + 1$

Hence $u(x) = u_c + u_p = ce^x + x + 1 = \frac{1}{y(x)}$

Therefore $y(x) = \frac{1}{ce^x + x + 1}$ is the general solution.

b) $\dot{y} - 2y = \sin xy^3$

c) $\dot{y} = xy^2 + y, y(0) = -3$ are exercise.

SECOND ORDER LINEAR DIFFERENTIAL WITH CONSTANT COEFFICIENTS

A linear second order differential equation with constant coefficients is any DE that can be written in the general form:

$$ay'' + by' + cy = f(x) \dots\dots (1)$$

If $f(x) \neq 0$, the equation (1) is called non-homogenous.

If $f(x) = 0$, we get

$$ay'' + by' + cy = 0 \dots\dots (2)$$

which is called associated homogenous or the reduced equation.

Example: 1. $2y'' - 3y' + 5y = 0$ is a homogeneous linear second-order differential equation.

2. $y'' + 6y = x^2 + 3$ is a non-homogeneous linear second-order differential equation.

Theorem: If $y = y_1(x)$ and $y = y_2(x)$ are two solutions to the homogenous DE (2), then

a) $y(x) = y_1(x) + y_2(x)$ is a solution to equation (2) and

b) $y(x) = cy_1(x)$ is a solution to equation (2) for any constant c .

Definition: A set of functions $\{y_1(x), y_2(x)\}$ is linearly independent (LI) if and only if for $\alpha_1, \alpha_2 \in \mathbb{R}$, $\alpha_1 y_1(x) + \alpha_2 y_2(x) = 0$ holds only when $\alpha_1 = \alpha_2 = 0$.

Definition: $\{y_1, y_2, \dots, y_n\}$ is linearly independent (LI) if and only if $\forall \alpha_1, \alpha_2, \dots, \alpha_n \in \mathbb{R}$,

$$\alpha_1 y_1 + \alpha_2 y_2 + \dots + \alpha_n y_n = 0 \Rightarrow \alpha_i = 0, \forall i = 1, 2, \dots, n$$

Example: Show that $\{e^x, e^{-x}\}$ is LI

Solution: Let $\alpha_1, \alpha_2 \in \mathbb{R}$ such that $\alpha_1 e^x + \alpha_2 e^{-x} = 0$ differentiating both sides we get

$$\Rightarrow \alpha_1 e^x - \alpha_2 e^{-x} = 0$$

In particular if $x = 0$

$$\begin{cases} \alpha_1 + \alpha_2 = 0 \\ \alpha_1 - \alpha_2 = 0 \end{cases} \Rightarrow \alpha_1 = \alpha_2 = 0$$

Therefore $\{e^x, e^{-x}\}$ is LI

Theorem: Criterion for linearly independent solutions

Let $y_1(x), y_2(x), \dots, y_n$ be $n - 1$ continuously differentiable functions on an interval I . Then the set of functions is linearly independent set on I if and only if the **Wronskian** of the functions,

$$W(y_1(x), y_2(x), \dots, y_n(x)) = \begin{vmatrix} y_1 & y_2 & \cdots & y_n \\ y_1' & y_2' & \cdots & y_n' \\ \vdots & \vdots & & \vdots \\ y_1^{n-1} & y_2^{n-1} & \cdots & y_n^{n-1} \end{vmatrix} \neq 0, \text{ for every } x \text{ in the interval } I.$$

In particular: if $n = 2$, $w(x) = \begin{vmatrix} y_1 & y_2 \\ \dot{y}_1 & \dot{y}_2 \end{vmatrix} = y_1 \dot{y}_2 - \dot{y}_1 y_2$.

Example: $w(e^x, e^{3x}) = \begin{vmatrix} e^x & e^{3x} \\ e^x & 3e^{3x} \end{vmatrix} = 2e^{4x}$

Theorem: If $\{y_1, y_2, \dots, y_n\}$ is linearly dependent, then $w(x) = 0$ but not the converse.

Theorem: The DE of (2) has two LI solutions.

Theorem: Any two solutions of (2) are LI if and only if $w(x) \neq 0$.

Definition: The set $\{y_1, y_2\}$ forms fundamental solution of (2) iff $\{y_1, y_2\}$ is LI Solution of (2).

Then $y_c(x) = c_1 y_1(x) + c_2 y_2(x)$ is the general solution of (2) or complementary solution of (2).

Example: In a DE $\ddot{y} - \frac{x}{x-1} \dot{y} + \frac{1}{x-1} y = 0$ on $(-\infty, 1) \cup (1, \infty)$

Take $y_1 = x, y_2 = e^x$ are solutions of the DE and are LI

Thus $\{x, e^x\}$ forms fundamental solution

Hence $y_c(x) = c_1 x + c_2 e^x$ is the general solution.

Theorem: If y_c is complementary solution and y_p is any solution of (1), then $y(x) = y_c + y_p$ is general

Solution of (1)

Example: In a DE $\ddot{y} + y = x^2$

Then the RHDE is $y' + y = 0$. Check that, $y_1 = \sin x, y_2 = \cos x$ are LI solutions.

And $y_p = x^2 - 2$ is a solution of the NHDE (by inspection)

Therefore, $y(x) = c_1 \sin x + c_2 \cos x + x^2 - 2$ is the general solution of the DE.

How to get y_1, y_2 , and y_p ?

Method of Solutions (finding y_1, y_2):

A linear second order differential equation with constant coefficients is any DE that can be written in the standard form

$$y'' + ay' + by = f(x) \dots\dots (1)$$

If $f(x) \neq 0$, the equation (1) is called non-homogenous.

If $f(x) = 0$, we get

$$y'' + ay' + by = 0 \dots\dots (2)$$

which is called associated homogenous or the reduced equation.

Suppose that a possible solution of (2) is $y = e^{mx}$. Since, $\dot{y} = me^{mx}$, $\ddot{y} = m^2 e^{mx}$, equation (2) takes the form

$$m^2 e^{mx} + ame^{mx} + be^{mx} = 0 \dots\dots\dots (3). \text{ As } e^{mx} \neq 0 \text{ for all } m \text{ and } x, \text{ we can divide equation (3)}$$

$$\text{by it to get } m^2 + am + b = 0 \dots\dots\dots (4) \text{ is called the characteristics equation of Eq. (2)}$$

Now, each value of m for which Eq. (4) holds will make $y = e^{mx}$ a solution of Eq. (2)

From Eq. (4), we have, $m = \frac{-a \pm \sqrt{a^2 - 4b}}{2}$ - characteristic root of Eq. (1). Here the value of m depends

on the sign of the expression under the square root.

Case I: $a^2 - 4b > 0$, $m^2 + am + b = 0$ has two distinct real roots. That is,

$$m_1 = \frac{-a + \sqrt{a^2 - 4b}}{2}, m_2 = \frac{-a - \sqrt{a^2 - 4b}}{2}$$

$y_1 = e^{m_1 x}$, $y_2 = e^{m_2 x}$ are distinct solutions (linearly independent.)

That is, $\{y_1 = e^{m_1 x}, y_2 = e^{m_2 x}\}$ - form a fundamental system solution.

$\Rightarrow y(x) = C_1 e^{m_1 x} + C_2 e^{m_2 x}$ - is the general solution of (2)

Case II: $a^2 - 4b = 0$. The equation $m^2 + am + b = 0$ has only one real (double) root.

$m_0 = \frac{-a}{2} \Rightarrow y_1(x) = e^{m_0 x} = e^{\frac{-a}{2}x}$ is the solution of (2). To find y_2 so that $\{y_1, y_2\}$ is linearly independent

$$y_2(x) = y_1(x) \int \frac{e^{-\int a dx}}{y_1^2(x)} dx = e^{\frac{-a}{2}x} \int \frac{e^{-ax}}{e^{-ax}} dx = x e^{\frac{-a}{2}x} = x y_1(x)$$

Therefore, $y_1(x) = e^{\frac{-a}{2}x}$ & $y_2(x) = x e^{\frac{-a}{2}x}$ are linearly independent solutions of (2). That is,

$\{y_1(x) = e^{m_0 x}, y_2(x) = x e^{m_0 x}\}$ - form a fundamental system of solutions.

$\Rightarrow y(x) = C_1 e^{m_0 x} + C_2 x e^{m_0 x}$ is the general solution of (2)

Case III: $a^2 - 4b < 0$. The equation $m^2 + am + b = 0$ has no real root that is, it has two complex roots.

$$m_{1,2} = \frac{-a \pm \sqrt{a^2 - 4b}}{2} = \frac{-a \pm i\sqrt{4b - a^2}}{2}, 4b - a^2 > 0$$

$$= \underbrace{\frac{-a}{2}}_{\alpha} \pm i \underbrace{\frac{\sqrt{4b - a^2}}{2}}_{\beta}, \text{ where } i = \sqrt{-1}$$

Therefore, the roots of $m^2 + am + b = 0$ are $m_1 = \alpha + \beta i$ & $m_2 = \alpha - \beta i$, m_1 & m_2 are conjugates.

Thus, from the conjugate roots: $m_1 = \alpha + \beta i$ & $m_2 = \alpha - \beta i$ of $m^2 + am + b = 0$,

$y_1(x) = e^{(\alpha + \beta i)x}$ & $y_2(x) = e^{(\alpha - \beta i)x}$ are solution of (2). These are complex solution for our real problem.

By Euler's formula, $y_1(x) = e^{(\alpha + \beta i)x} = e^{\alpha x} (\cos \beta x + i \sin \beta x)$ & $y_2(x) = e^{(\alpha - \beta i)x} = e^{\alpha x} (\cos \beta x - i \sin \beta x)$

$$\text{Put } y_1^*(x) = \frac{1}{2}(y_1 + y_2) \text{ \& } y_2^*(x) = \frac{1}{2}(y_2 - y_1), \quad y_1^*(x) = e^{\alpha x} \cos \beta x \text{ \& } y_2^* = e^{\alpha x} \sin \beta x$$

Therefore, y_1^* & y_2^* are real solutions of (2). $y_c(x) = C_1 e^{\alpha x} \cos \beta x + C_2 e^{\alpha x} \sin \beta x$ - is the general solution of (2). Now, collecting all these things together, we have the following result.

Theorem: Given the homogeneous second order linear differential equation with constant coefficients:

$$\ddot{y} + a\dot{y} + by = 0 \text{ ----- (2)}$$

i) If $a^2 - 4b > 0$, then $\{y_1 = e^{m_1 x}, y_2 = e^{m_2 x}\}$ is a fundamental system of solutions and

$$y_c(x) = C_1 e^{m_1 x} + C_2 e^{m_2 x} \text{ - is the general solution of (2)}$$

ii) If $a^2 - 4b = 0$, then $\{y_1(x) = e^{m_0 x}, y_2(x) = x e^{m_0 x}\}$ is a fundamental system of solutions.

$$y_c(x) = C_1 e^{m_0 x} + C_2 x e^{m_0 x} \text{ is the general solution of (2).}$$

iii) If $a^2 - 4b < 0$, then $\{e^{\alpha x} \cos \beta x, e^{\alpha x} \sin \beta x\}$ is a fundamental system of solutions, where

$$m_{1,2} = \alpha \pm \beta i \text{ are complex solutions of } m^2 + am + b = 0.$$

Thus, $y_c(x) = C_1 e^{\alpha x} \cos \beta x + C_2 e^{\alpha x} \sin \beta x$ is the general solution of (2).

In general, if y_1, y_2 are linearly independent solutions of (2) on I , then $y_c = C_1 y_1 + C_2 y_2$ is the complementary solution of (2), and if y_p is any one solution of (1) on I , then

$y(x) = y_c + y_p = C_1 y_1 + C_2 y_2 + y_p$ is the general solution of (1) on I , where C_1 and C_2 are constants.

Example: a) Solve $\ddot{y} - 2\dot{y} = 0$

Solution: Trial solution $y = e^{mx}$

Characteristic equation: $m^2 - 2m = 0 \Rightarrow m = 0$ or $m = 2$, $y_1(x) = e^{0x} = 1$ & $y_2(x) = e^{2x}$ are linearly independent solution. So $\{1, e^{2x}\}$ form fundamental system of solutions. The general solution of $\ddot{y} - 2\dot{y} = 0$ is $y_c(x) = C_1 + C_2 e^{2x}$.

b) Solve $4\ddot{y} - 4\dot{y} + y = 0$

Solution: Trial solution $y = e^{mx}$

Characteristic equation: $4m^2 - 4m + 1 = 0 \Rightarrow m = \frac{1}{2}$ double root

Therefore, $y_1(x) = e^{\frac{1}{2}x}$ & $y_2(x) = xe^{\frac{1}{2}x}$ are linearly independent solutions.

Thus, $y_c(x) = e^{\frac{1}{2}x} (C_1 + C_2 x)$ is the general solution.

c) Solve $\ddot{y} + 4y = 0$

Solution: Trial solution $y = e^{mx}$

Characteristic equation: $m^2 + 4 = 0 \Rightarrow m = \pm 2i$, complex roots; $\alpha = 0, \beta = 2$

Therefore, $y_1(x) = e^{\alpha x} \cos \beta x = \cos 2x$, & $y_2(x) = e^{\alpha x} \sin \beta x = \sin 2x$ are linearly independent solutions.

Thus, $y_c(x) = C_1 \cos 2x + C_2 \sin 2x$ - is the general solution.

d) Solve $\ddot{y} - 3\dot{y} + 2y = 2e^{3x}$, $y(0) = 1, \dot{y}(0) = 0$ (nonhomogeneous) ----- (*)

Solution: RHDE: $\ddot{y} - 3\dot{y} + 2y = 0$ ----- (**). This is homogenous 2nd order differential equation with Constant coefficients.

Characteristic equation: $m^2 - 3m + 2 = 0 \Rightarrow (m-1)(m-2) = 0 \Rightarrow m = 1$ or $m = 2$.

Therefore, $y_1(x) = e^x$ & $y_2(x) = e^{2x}$ are linearly independent solutions.

$y_c(x) = C_1 e^x + C_2 e^{2x}$ - is the general solution of (*).

Finding Particular solution $y_p(x)$

THE METHOD OF UNDETERMINED COEFFICIENTS

In this section we present a method for finding a particular solution $y_p(x)$ to the non-homogenous linear differential equations: $\ddot{y} + a\dot{y} + by = f(x)$ ----- (1), where a and b are constants.

And, the method requires that the function $f(x)$ in the equation (1) be one of the following three forms:

i) $p_k(x)$, polynomial of degree k

ii) $p_k(x)e^{\alpha x}$

iii) $p_k(x)e^{\alpha x} \sin \beta x + p_k(x)e^{\beta x} \cos \beta x$, and the method fails if $f(x)$ is different from these three forms.

- ❖ The method of undetermined coefficients assumes that the solution to equation (1) is of exactly the same form as $f(x)$.
- ❖ The techniques require that we replace each dependent variable y in (1) with an expression of the same form as $f(x)$ having polynomial terms with undetermined coefficients. If we compare both sides of the resulting equation, it is possible to determine the unknown coefficients.

To illustrate this method, we consider a number of examples.

Example: 1. Solve the differential equation $\ddot{y} - y = x^2$ ----- (1)

Solution: since $f(x) = x^2$ is the polynomial of degree two, we guess that (1) has a particular solution $y_p(x)$ that is a polynomial of degree two. We try the most general polynomial of degree two. That is, $y_p(x) = ax^2 + bx + c$, where a, b and c are constants to be determined from the given differential equation. Then $\dot{y}_p = 2ax + b$ & $\ddot{y}_p = 2a$, so if we substitute $\ddot{y}_p$ & y_p in to (1), we obtain

$$2a - (ax^2 + bx + c) = x^2$$

Equating coefficients of like powers of the variable x , we have $2a - c = 0, -b = 0, -a = 1$,

$\Rightarrow a = -1, b = 0, c = 2a \Rightarrow c = 2(-1) = -2$, and the particular solution $y_p(x) = -x^2 - 2$ and

RHDE: $\ddot{y} - y = 0$

Characteristic equation: $m^2 - 1 = 0 \Rightarrow m_1 = \pm 1$. Therefore, $y_c(x) = c_1 e^{-x} + c_2 e^x$ is the complementary solution of $\ddot{y} - y = 0$.

Thus, $y = y_c + y_p = C_1 e^{-x} + C_2 e^x - x^2 - 2$ is the general solution of the differential equation $\ddot{y} - y = x^2$.

2. Solve $\ddot{y} + y = xe^{2x}$

Solution: Here $f(x)$ is of the form (ii), where $p_k(x)$ is a polynomial of degree one, so we try of solution of the form $y_p(x) = e^{2x}(a + bx)$. Then $\dot{y}_p(x) = e^{2x}(2a + 2bx + b)$, $\ddot{y}_p(x) = e^{2x}(4a + 4bx + 4b)$, and substitution

yields, $e^{2x}(4a+4bx+4b)+e^{2x}(a+bx)=xe^{2x}$ and equating like powers of x , we obtain the equations

$$5a+4b=0, 5b=1 \Rightarrow a=-\frac{4}{25}, b=\frac{1}{5}, \text{ and the particular solution is } y_p(x)=\frac{e^{2x}}{25}(5x-4) \text{ and}$$

$$\text{RHDE: } \ddot{y}+y=0$$

$$\text{Characteristic equation: } m^2+1=0 \Rightarrow m=\pm 1i, \text{ complex roots; } \alpha=0, \beta=1$$

Therefore, $y_1(x)=e^{\alpha x} \cos \beta x = \cos x$, & $y_2(x)=e^{\alpha x} \sin \beta x = \sin x$ are linearly independent solutions.

Thus, $y_c(x)=C_1 \cos x + C_2 \sin x$ is the complementary solution of $\ddot{y}+y=0$.

Hence, $y=y_c+y_p=C_1 \cos x + C_2 \sin x + \frac{e^{2x}}{25}(5x-4)$ is the general solution of the differential equation

$$\ddot{y}+y=xe^{2x}$$

Theorem 1:

Consider the DE $\ddot{y}+a\dot{y}+by=p(x)$ ----- (1), where a and b are constants, $p_k(x)$, polynomial of degree k .

- i) If $m=0$ is not a root of the characteristics equation of the RHDE, then there exists a particular solution to equation (1) of the form $y_p=A_k x^k + \dots + A_1 x + A_0$
- ii) If $m=0$ is a simple root of the characteristics equation of the RHDE, then there exists a particular solution to equation (1) of the form $y_p=x(A_0 + A_1 x + \dots + A_k x^k)$
- iii) If $m=0$ is a repeated root of the characteristics equation of the RHDE, then there exists a particular solution to equation (1) of the form $y_p=x^2(A_0 + A_1 x + \dots + A_k x^k)$

Example: 1. Solve $\ddot{y}+4\dot{y}+3y=6x$ ----- (1)

$$\text{Solution: RHDE: } \ddot{y}+4\dot{y}+3y=0$$

$$\text{Characteristic equation: } m^2+4m+3=0 \Rightarrow (m+1)(m+3)=0 \Rightarrow m=-1, m=-3.$$

Therefore, $y_c(x)=c_1 e^{-x} + c_2 e^{-3x}$ is the complementary solution of $\ddot{y}+4\dot{y}+3y=0$.

To find y_p by the method of undetermined coefficient, first we need to check that zero (0) is the root of the characteristic equation $m^2+4m+3=0 \Rightarrow 0^2+4(0)+3=3 \neq 0$. Therefore, 0 is not a root.

Therefore, $y_p(x)$ takes the form: $y_p(x)=A_1 x + A_0 \Rightarrow \dot{y}_p=A_1, \ddot{y}_p=0$, substituting in (1), we have,

$$0+4A_1+3(A_1 x + A_0)=6x \Rightarrow (4A_1+3A_0)+3A_1 x=6x \Rightarrow \begin{cases} 4A_1+3A_0=0 \Rightarrow A_0=-\frac{4A_1}{3} \\ 3A_1=6 \Rightarrow A_1=2 \end{cases}$$

Thus, $y_p(x) = 2x - \frac{8}{3}$.

Hence, the general solution of the differential equation is $y(x) = c_1 e^{-x} + c_2 e^{-3x} + 2x - \frac{8}{3}$.

Example: 2. Solve $\ddot{y} + 5\dot{y} = 4x^2$ ----- (1)

Solution: RHDE: $\ddot{y} + 5\dot{y} = 0$

Characteristic equation is $m^2 + 5m = 0$, with roots $m = 0, m = -5$. Consequently,

$y_c(x) = c_1 + c_2 e^{-5x}$ is the complementary solution of $\ddot{y} + 5\dot{y} = 0$.

The usual trial solution corresponding to the driving term (function) $f(x) = 4x^2$ is

$y_p(x) = A_0 + A_1 x + A_2 x^2$. However, since $m = 0$ a simple root of the characteristics equation of the RHDE, we must use the modified trial solution $y_p(x) = x(A_0 + A_1 x + A_2 x^2)$.

Inserting this expression into equation (1) yields

$$\begin{aligned} 2A_1 + 6A_2 x + 5(A_0 + 2A_1 x + 3A_2 x^2) &= 4x^2 \\ \Rightarrow 5A_0 + 2A_1 + (10A_1 + 6A_2)x + 15A_2 x^2 &= 4x^2 \end{aligned}$$

Consequently, the coefficients must be chosen to satisfy

$$5A_0 + 2A_1 = 0$$

$$10A_1 + 6A_2 = 0$$

$$15A_2 = 4$$

$$\begin{cases} A_2 = \frac{4}{15} \\ A_1 = -\frac{4}{15} \\ A_0 = \frac{8}{125} \end{cases}, \text{ so that a particular solution to equation (1) is}$$

$y_p(x) = \frac{4}{375} x(6 - 15x + 25x^2)$. The general solution to the DE is therefore

$$y(x) = c_1 + c_2 e^{-5x} + \frac{4}{375} x(6 - 15x + 25x^2).$$

Exercise: solve the differential equation: a) $\ddot{y} - \dot{y} = x^2 + 5$

$$\text{b) } \ddot{y} + 2\dot{y} + 2y = 3x^2$$

Theorem 2: Consider the DE $\ddot{y} + a\dot{y} + by = A \cos \beta x + B \sin \beta x$ ----- (1), where a, b, A , and β are constants.

- i) If $m = \beta i$ is not a root of the characteristics equation of the RHDE, then there exists a particular solution to equation (1) of the form $y_p(x) = A_0 \cos \beta x + B_0 \sin \beta x$.
- ii) If $m = \beta i$ is a root of the characteristics equation of the RHDE, then there exists a particular solution to equation (1) of the form $y_p(x) = x(A_0 \cos \beta x + B_0 \sin \beta x)$.

Example: Solve the $\ddot{y} - \dot{y} - 2y = 20\sin x$ ----- (1)

Solution: RHDE: $\ddot{y} - \dot{y} - 2y = 0$ ----- (2)

Characteristic equation: $m^2 - m - 2 = 0 \Rightarrow (m - 2)(m + 1) = 0 \Rightarrow m = 2, m = -1$

So that $y_c(x) = c_1 e^{2x} + c_2 e^{-x}$. In this case, an appropriate trial solution is $y_p(x) = A_0 \cos x + A_1 \sin x$.

Substituting this trial solution into equation (1) yields $(A_0 - 3A_1)\sin x - (3A_0 + A_1)\cos x = 20\sin x$

This equation is satisfied for all x if and if $\begin{cases} -3A_0 - A_1 = 0 \\ A_0 - 3A_1 = 20 \end{cases} \Rightarrow \begin{cases} A_1 = -6 \\ A_0 = 2 \end{cases}$. So that a particular solution to equation (1) is $y_p(x) = 2\cos x - 6\sin x$.

Consequently, (2) has general solution $y(x) = c_1 e^{2x} + c_2 e^{-x} + 2\cos x - 6\sin x$

Exercise: solve the DE 1) $\ddot{y} + 16y = 24\cos 4x$

$$2) \ddot{y} + 4\dot{y} + 5y = 24\sin x$$

Theorem 3: Consider the DE $\ddot{y} + a\dot{y} + by = p(x)e^{\alpha x}$ ----- (1), where a, b , and α are constants, $p_k(x)$, polynomial of degree k .

- i) If $m = \alpha$ is not a root of the characteristics equation of the RHDE, then there exists a particular solution to equation (1) of the form $y_p(x) = (A_0 + A_1 x + \dots + A_k x^k)e^{\alpha x}$
- ii) If $m = \alpha$ is a simple root of the characteristics equation of the RHDE, then there exists a particular solution to equation (1) of the form $y_p(x) = x(A_0 + A_1 x + \dots + A_k x^k)e^{\alpha x}$
- iii) If $m = \alpha$ is a repeated root of the characteristics equation of the RHDE, then there exists a particular solution to equation (1) of the form $y_p = x^2(A_0 + A_1 x + \dots + A_k x^k)e^{\alpha x}$

Example: Find the particular solution y_p for the differential equation $\ddot{y} - 2\dot{y} = xe^{2x}$

Solution: characteristic equation $m^2 - 2m = 0 \Rightarrow m = 0, m = 2$

Here $p_k(x)e^{\alpha x} = xe^{2x} \Rightarrow \alpha = 2, \beta = 0$ and $\alpha = 2$ is a root of $m^2 - 2m = 0$

y_p must have the form: $y_p(x) = x(A_1 x + A_0)e^{2x}$, and determine A & B by substitution in the differential equation.

Exercise: Solve the DE 1) $\ddot{y} + 4\dot{y} + 4y = 10e^{-2x}$

$$2) \ddot{y} - y = 9xe^{3x}, y(0) = 0, \dot{y}(0) = 9$$

Theorem 4: Consider the DE $\ddot{y} + a\dot{y} + by = p(x)e^{\alpha x} A \cos \beta x$ or $p(x)e^{\alpha x} B \sin \beta x$ ----- (1), where a, b, α , and β are constants, $p_k(x)$, polynomial of degree k .

- i) If $m = \alpha + \beta i$ is not a root of the characteristics equation of the RHDE, then there exists a particular solution to equation (1) of the form

$$y_p(x) = (A_0 + A_1x + \dots + A_kx^k)e^{\alpha x} \cos \beta x + (B_0 + B_1x + \dots + B_kx^k)e^{\alpha x} \sin \beta x$$

- ii) If $m = \alpha + \beta i$ is a simple root of the characteristics equation of the RHDE, then there exists a particular solution to equation (1) of the form

$$y_p(x) = x[(A_0 + A_1x + \dots + A_kx^k)e^{\alpha x} \cos \beta x + (B_0 + B_1x + \dots + B_kx^k)e^{\alpha x} \sin \beta x]$$

Example: Solve $\ddot{y} + 4y = 16x \cos 2x$ ----- (1)

Solution: RHDE: $\ddot{y} + 4y = 0$ ----- (2)

Characteristic equation: $m^2 + 4 = 0 \Rightarrow m = \pm 2i$. Here, $\alpha = 0, \beta = 2 \Rightarrow m = \alpha + \beta i = 0 + 2i = 2i$ is a simple root of the characteristic equation of (2). And an appropriate trial solution is

$$y_p(x) = x[(A_0 + A_1x) \cos 2x + (B_0 + B_1x) \sin 2x].$$

$$\Rightarrow y_p(x) = 2x^2 \sin 2x + x \cos 2x, \text{ and complementary solution of (2) is } y_c(x) = c_1 \sin 2x + c_2 \cos 2x$$

Therefore, the general solution of (1) is $y(x) = y_c(x) + y_p(x) = (c_1 + 2x^2) \sin 2x + (c_2 + x) \cos 2x$.

Theorem 5: Consider the DE $\ddot{y} + a\dot{y} + by = f(x)$, where a , and b are constants. If $f(x)$ is a sum of terms of the form considered in **theorems 1-4**, then the appropriate trial form of a particular solution is obtained by taking the sum of the corresponding trial solution.

Example: 1. Determine an appropriate trial form of a particular solution of

$$\text{a) } \ddot{y} + 2\dot{y} = 3e^{-2x} + 4x^2 \quad \text{b) } \ddot{y} - 4y = 2 \cos x + 5xe^{2x}$$

Solution: (a) The characteristic equation is $m^2 + 2m = 0$, with roots $m = 0, m = -2$. Consequently, the complementary solution is $y_c(x) = c_1 + c_2 e^{-2x}$

Therefore, a trial solution corresponding to the driving term $f_1(x) = 3e^{-2x}$ is $y_{p_1}(x) = B_0 x e^{-2x}$, since $m = \alpha = -2$ is a simple root of the characteristic equation of the RHDE.

Whereas a trial solution corresponding to the driving term $f_2(x) = 4x^2$ is $y_{p_2}(x) = x(A_0 + A_1x + A_2x^2)$, since $m = \alpha = 0$ is a simple root of the characteristic equation of the RHDE.

Hence, an appropriate trial solution for the given DE is $y_p(x) = y_{p_1} + y_{p_2} = x(A_0 + A_1x + A_2x^2) + B_0xe^{-2x}$

b) The characteristic equation is $m^2 - 4 = 0$, with roots $m = \pm 2$. Consequently, the complementary solution is $y_c(x) = c_1e^{2x} + c_2e^{-2x}$

Therefore, a trial solution corresponding to the driving term $f_1(x) = 5xe^{2x}$ is $y_{p_1}(x) = x(B_0 + B_1)e^{2x}$, since $m = \alpha = 2$ is a simple root of the characteristic equation of the RHDE.

Whereas a trial solution corresponding to the driving term $f_2(x) = 2\cos x$ is $y_{p_2}(x) = A_0\cos x + A_1\sin x$, since $m = \beta i = i$ is not a root of the characteristic equation of the RHDE.

Hence, an appropriate trial solution for the given DE is

$$y_p(x) = y_{p_1} + y_{p_2} = x(B_0 + B_1x)e^{2x} + A_0\cos x + A_1\sin x$$

Example: 2. Solve the differential equation $\ddot{y} - 4\dot{y} + 4y = e^x - x$

Solution: RHDE: $\ddot{y} - 4\dot{y} + 4y = 0$

Characteristic equation: $m^2 - 4m + 4 = 0 \Leftrightarrow (m - 2)^2 = 0 \Rightarrow m = 2$, double root.

Fundamental system of solution is $\{e^{2x}, xe^{2x}\}$, $\Rightarrow y_c(x) = c_1e^{2x} + c_2xe^{2x}$, complementary solution.

Particular solution: y_p ?, $f(x) = -x + e^x$, and $f(x) = g(x) + h(x)$, where $\underbrace{g(x) = -x}_{\alpha=0}, \underbrace{h(x) = e^x}_{\alpha=1}$

$y_{pf} = y_{pg} + y_{ph}$, and 0 & 1 are not roots of the characteristic equation $m^2 - 4m + 4 = 0$

Therefore, $y_p = y_{p_1} + y_{p_2}$ (by principle of superposition)

$$= (A_1x + A_0) + Be^x. \text{ Find } A_1, A_0 \text{ \& } B \text{ (exercise)}$$

Exercise: Solve the following differential equation by the method of undetermined coefficients:

a) $\ddot{y} + 3\dot{y} + 2y = 12e^x$

b) $\ddot{y} - 3\dot{y} = 2e^{2x} \sin x$

c) $\ddot{y} - 3\dot{y} + 2y = 2x^2 + 3e^x$

d) $\ddot{y} + 2\dot{y} + y = (x - 1) + xe^{-x}$

METHOD OF VARIATION OF PARAMETERS

-is a method for finding the particular solution to non-homogenous differential equation

Suppose $\{y_1, y_2\}$ forms a fundamental solution of (2) (i.e., y_1 , and y_2 are known), then the

complementary solution of (2) is $y_p(x) = c_1y_1(x) + c_2y_2(x)$. To find a particular solution of $y_p(x)$ of (1),

vary the constants c_1 and c_2 to functions of x .

Therefore, $y_p(x) = c_1(x)y_1(x) + c_2(x)y_2(x)$ (trial form), then $\dot{y}_p = \dot{c}_1 y_1 + c_1 \dot{y}_1 + \dot{c}_2 y_2 + c_2 \dot{y}_2$

Substituting y_p in the DE $\ddot{y}_p + a(x)\dot{y}_p + b(x)y_p = f(x)$ we have

$$\begin{cases} \dot{c}_1 y_1 + \dot{c}_2 y_2 = 0 \\ \dot{c}_1 \dot{y}_1 + \dot{c}_2 \dot{y}_2 = f(x) \end{cases} \Leftrightarrow \begin{bmatrix} \dot{c}_1 \\ \dot{c}_2 \end{bmatrix} \begin{bmatrix} y_1 & y_2 \\ \dot{y}_1 & \dot{y}_2 \end{bmatrix} = \begin{bmatrix} 0 \\ f \end{bmatrix}$$

Since $w(x) = \begin{vmatrix} y_1 & y_2 \\ \dot{y}_1 & \dot{y}_2 \end{vmatrix} \neq 0$

Using crammer's rule $\dot{c}_1(x) = \frac{\begin{vmatrix} 0 & y_2 \\ f & \dot{y}_2 \end{vmatrix}}{w(x)} = \frac{-f(x)y_2}{w(x)}$ and $\dot{c}_2(x) = \frac{\begin{vmatrix} y_1 & 0 \\ y_2 & f \end{vmatrix}}{w(x)} = \frac{f(x)y_1}{w(x)}$

$$\Rightarrow c_1(x) = \int \frac{-f(x)y_2}{w(x)} dx \text{ and } c_2(x) = \int \frac{f(x)y_1}{w(x)} dx$$

d) Solve $\ddot{y} - 3\dot{y} + 2y = 2e^{3x}$, $y(0) = 1, \dot{y}(0) = 0$ (non homogenous) ----- (*)

Solution: RHDE: $\ddot{y} - 3\dot{y} + 2y = 0$ ----- (**). This is homogenous 2nd order differential equation with Constant coefficients.

Characteristic equation: $m^2 - 3m + 2 = 0 \Rightarrow (m-1)(m-2) = 0 \Rightarrow m = 1 \text{ or } m = 2$.

Therefore, $y_1(x) = e^x$ & $y_2(x) = e^{2x}$ are linearly independent solutions.

$y_c(x) = C_1 e^x + C_2 e^{2x}$ - is the general solution of (*).

Particular solution, $y_p(x) = C_1(x)y_1(x) + C_2(x)y_2(x)$ (By using variation of parameter) and

$$w(e^x, e^{2x}) = \begin{vmatrix} e^x & e^{2x} \\ e^x & 2e^{2x} \end{vmatrix} = e^{3x}$$

$$\dot{C}_1(x) = \frac{\begin{vmatrix} 0 & e^{2x} \\ f(x) & 2e^{2x} \end{vmatrix}}{w(x)} = \frac{-e^{2x} f(x)}{w(x)} = \frac{-e^{2x} 2e^{3x}}{e^{3x}} = -2e^{2x} \Rightarrow C_1(x) = -2 \int e^{2x} dx = -e^{2x}$$

$$\dot{C}_2(x) = \frac{\begin{vmatrix} e^x & 0 \\ e^x & 2e^{3x} \end{vmatrix}}{w(x)} = \frac{2e^x \cdot e^{3x}}{e^{3x}} = 2e^x \Rightarrow C_2(x) = \int 2e^x dx = 2e^x$$

Therefore, $y_p(x) = -e^{3x} + 2e^{3x} = e^{3x}$

Thus, $y = y_c + y_p = C_1 e^x + C_2 e^{2x} + e^{3x}$ is the general solution of the differential equation (*).

To find the unique solution satisfying the given initial conditions:

$$y(0)=1 \Rightarrow y(0)=C_1+C_2+1=1 \Rightarrow C_1+C_2=0 \Rightarrow C_1=-C_2 \text{-----(1)}$$

$$\dot{y}(x)=C_1e^x+2C_2e^{2x}+3e^{3x} \Rightarrow \dot{y}(0)=C_1+2C_2+3=0 \Rightarrow C_1+2C_2=-3 \text{-----(2)}$$

From, (1) and (2), we have $C_1=3, C_2=-3$

Therefore, the unique solution of the IVP is $y=3e^x-3e^{2x}+e^{3x}$

Exercise: Solve the following differential equations:

a) $\ddot{y}+4y=\sin x$

b) $\ddot{y}+\dot{y}-6y=10e^{3x}, y(0)=0, \dot{y}(0)=3$

c) $\ddot{y}+y=3x^2, y(0)=4, \dot{y}(0)=0$

d) $\ddot{y}-4\dot{y}+4y=6xe^{3x}, y(0)=0, \dot{y}(0)=3$

CHAPTER TWO: LAPLACE TRANSFORM

Definition: Let $f(t)$ be function defined on the interval $[0, \infty)$. Then the Laplace transform of f is the

function $F(s)=\int_0^{\infty} e^{-st} f(t)dt$, provided that the improper integral converges, and usually denoted by

$L(f(t))=L[f]=L\{f(t)\}$. Here L is Laplarian operator

Example: Find a) $L\{e^{at}\}$

$$\text{Solution: } L\{e^{at}\} = \int_0^{\infty} e^{-st} e^{at} dt = \int_0^{\infty} e^{t(s-a)} dt = \lim_{T \rightarrow \infty} \int_0^T e^{t(s-a)} dt = \begin{cases} \infty, s \leq a \\ \frac{1}{s-a}, s > a \end{cases}$$

$$\text{Therefore } L\{e^{at}\} = \frac{1}{s-a}, s > a$$

$$\text{In particular } L\{e^t\} = \frac{1}{s-1}, s > 1 \text{ and } L\{e^{2t} - 3e^{-t}\} = L\{e^{2t}\} - 3L\{e^{-t}\} = \frac{1}{s-2} - 3\frac{1}{s+1}, s > 2 \text{ and } s > -1$$

$$\Rightarrow L\{e^{2t} - 3e^{-t}\} = \frac{1}{s-2} - 3\frac{1}{s+1} = \frac{s+1-3s+6}{(s-2)(s+1)} = \frac{-2s+7}{(s-2)(s+1)}, s > 2$$

$$\text{For } a=0, L\{1\} = \frac{1}{s}, s > 0$$

b) $L\{\cos at\}, L\{\sin at\}$

$$\text{Solution: } L\{e^{iat}\} = \frac{1}{s-ia} = \frac{1}{s-ia} \frac{s+ia}{s+ia} = \frac{s+ia}{s^2+a^2} = \frac{s}{s^2+a^2} + i \frac{a}{s^2+a^2}$$

However, from Euler's formula $e^{iat} = \cos at + i \sin at$

$$L\{e^{iat}\} = L\{\cos at + i \sin at\} = L\{\cos at\} + L\{i \sin at\} = \frac{s}{s^2 + a^2} + i \frac{a}{s^2 + a^2}$$

Thus equating the real and imaginary parts we have

$$L\{\cos at\} = \frac{s}{s^2 + a^2}, s > 0, \text{ and } L\{\sin at\} = \frac{a}{s^2 + a^2}, s > 0$$

Example: $L\{\cos 3t\} = \frac{s}{s^2 + 9}, s > 0$ and $L\{\sin -3t\} = \frac{-3}{s^2 + 9}, s > 0$

Theorem: for $n = 0, 1, 2, \dots$ $L\{t^n\} = \frac{n!}{s^{n+1}}, s > 0$ (by induction)

Example: $L\{t^0\} = \frac{0!}{s^{0+1}} = \frac{1}{s}, s > 0$, $L\{t^2\} = \frac{2!}{s^{2+1}} = \frac{2}{s^3}, s > 0$ and $L\{t^3\} = \frac{6}{s^4}, s > 0$

Theorem: $L\{c_1 f(t) + c_2 g(t)\} = c_1 L\{f(t)\} + c_2 L\{g(t)\}$ (L is linear)

Exercise: a) $L\{3e^{2t} - 4t^2\}$ b) $L\{2t^2 - 5\}$ c) $L\{3\sin 2t - 2t^4 + \cos 6t\}$

Solution: By above theorem, $L\{3e^{2t} - 4t^2\} = L\{3e^{2t}\} - L\{4t^2\} = 3L\{e^{2t}\} - 4L\{t^2\}$

$$= 3 \frac{1}{s-2} - 4 \frac{2!}{s^3} = \frac{3}{s-2} - \frac{8}{s^3}$$

b and c, exercise.

Example: If $f(t) = \begin{cases} t, & 0 \leq t < 2 \\ 3, & t \geq 2 \end{cases}$, the find $L\{f(t)\}$

Solution: $F(s) = L\{f(t)\} = \int_0^{\infty} f(t)e^{-st} dt = \int_0^2 te^{-st} dt + \int_2^{\infty} 3e^{-st} dt = \int_0^2 te^{-st} dt + \lim_{T \rightarrow \infty} \int_2^T 3e^{-st} dt$

But check that $\int_2^{\infty} e^{-st} dt = \frac{e^{-st}}{-s}, s > 0$

Therefore $F(s) = L\{f(t)\} = \int_0^{\infty} f(t)e^{-st} dt = \int_0^2 te^{-st} dt + \int_2^{\infty} 3e^{-st} dt = \frac{1}{s^2} + e^{-2s} \frac{s^2 - 1}{s^2}, s > 0$

Exercise: find $L\{f(t)\}$ for a) $f(t) = e^{at} + e^{-at}$ b) $f(t) = \begin{cases} 1, & 0 \leq t \leq 1 \\ 0, & t > 1 \end{cases}$

Theorem: (1st shifting theorem)

If $L\{f(t)\} = F(s)$, then

$$1) L\{e^{at} f(t)\} = F(s-a), s > a \text{ (s - shift theorem)}$$

$$2) L\{t^n f(t)\} = (-1)^n \frac{d^n}{ds^n} F(s) \text{ (t - shift theorem)}$$

Example: Find a) $L\{e^{-2t} \cos 3t\}$ b) $L\{te^{at}\}$ c) $L\{t^2 \sin t\}$

Solution: a) $F(s) = L\{\cos 3t\} = \frac{s}{s^2 + 9}, s > 0 \Rightarrow L\{e^{-2t} \cos 3t\} = F(s+2) = \frac{s+2}{(s+2)^2 + 9}, s > 2$

b) $F(s) = L\{e^{at}\} = \frac{1}{s-a}, s > a \Rightarrow L\{te^{at}\} = (-1)^1 \frac{d}{ds} F(s) = -\frac{d}{ds} \left(\frac{1}{s-a} \right) = \frac{1}{(s-a)^2}, s > a$ or

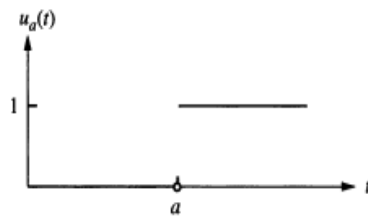
$$F(s) = L\{t\} = \frac{1}{s^2}, s > 0 \Rightarrow L\{e^{at}t\} = F(s-a) = \frac{1}{(s-a)^2}, s > a$$

c) $F(s) = L\{\sin t\} = \frac{1}{s^2 + 1}, s > 0 \Rightarrow L\{t^2 \sin t\} = (-1)^2 \frac{d^2}{ds^2} F(s) = \frac{d^2}{ds^2} \left(\frac{1}{s^2 + 1} \right) = \frac{d}{ds} \left(\frac{2s}{(s^2 + 1)^2} \right), s > 0$

Exercise: Find $L\{f(t)\}$ a) $f(t) = e^{3t} t^2 \sin 4t$ b) $f(t) = te^{-2t} \cos 3t$ c) $f(t) = \cos^2 t + \sinh 2t - \cosh -2t$

Definition :(unit step function)

The unit step function or Heaviside step function $u(t-a)$ or $u_a(t)$, is defined by $u(t-a) = \begin{cases} 0, & 0 \leq t < a \\ 1, & t \geq a \end{cases}$



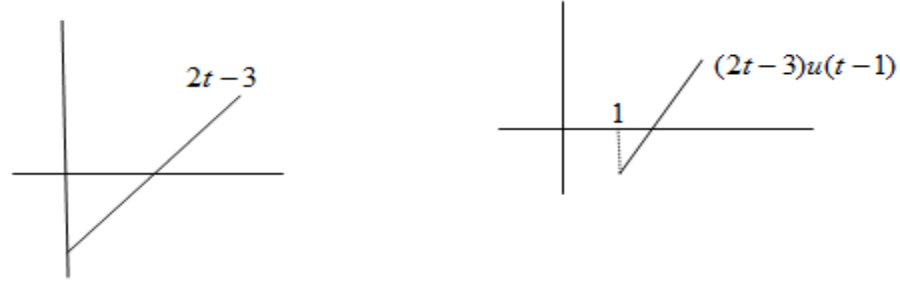
Graph of unit step function

-it is a switch that is turned on at $t = a$

It is used to - “turns off” a portion of the graph of a function f defined for $t \geq 0$

-write piecewise defined (step) function in compact form

Example: consider the function $f(t) = 2t - 3$. To “turn off” the portion the graph of f for $0 \leq t \leq 1$, we simply form the product $(2t - 3)u(t - 1)$.



A general piecewise defined function of the type $f(t) = \begin{cases} g(t), & 0 < t < a \\ h(t), & t \geq a \end{cases}$ is the same as

$$f(t) = g(t) - g(t)u(t-a) + h(t)u(t-a)$$

Similarly, a step function of the type $f(t) = \begin{cases} 0, & 0 \leq t < a \\ g(t), & a \leq t < b \\ 0, & t \geq b \end{cases}$ can be written $f(t) = g(t)[u(t-a) - u(t-b)]$

and a function of $f(t) = \begin{cases} g(t), & 0 \leq t < a \\ h(t), & a \leq t < b \\ k(t), & t \geq b \end{cases}$ can be written

$$f(t) = g(t) - g(t)u(t-a) + h(t)[u(t-a) - u(t-b)] + k(t)u(t-b)$$

Example: Express the step function $f(t) = \begin{cases} -t^2 + 3t - 2, & 1 \leq t < 2 \\ 0, & t \geq 2 \end{cases}$ in terms of unit step function.

Solution: $h(t) = -t^2 + 3t - 2$, $k(t) = 0$, and no $g(t)$

$$\begin{aligned} \Rightarrow f(t) &= -t^2 + 3t - 2[u(t-1) - u(t-2)] \\ &= \underbrace{-t^2 + 3t - 2(t-1)}_{\text{switch on at } t=1} - \underbrace{(-t^2 + 3t - 2)u(t-2)}_{\text{switch off at } t=2} \end{aligned}$$

Theorem (second shifting theorem):

Let $L\{f(t)\} = F(s)$. Then, $L\{u(t-a)f(t-a)\} = e^{-as}F(s)$

Corollary: Let $L\{f(t)\} = F(s)$. Then $L\{u(t-a)f(t)\} = e^{-as}L\{f(t+a)\}$

Example: If $f(t) = \begin{cases} 0, & 0 \leq t < \pi \\ 3\cos t, & t \geq \pi \end{cases}$, then find $L\{f(t)\}$

Solution: $g(t) = 0$, $h(t) = 3\cos t$, $a = \pi$

$$\begin{aligned} \Rightarrow f(t) &= 0 - 0u(t-\pi) + 3\cos t u(t-\pi) \\ \Rightarrow f(t) &= 3\cos t u(t-\pi) \end{aligned}$$

Therefore, $L\{f(t)\} = L\{3\cos t u(t - \pi)\} = 3e^{-\pi s} L\{\cos(t + \pi)\} = -3e^{-\pi s} L\{\cos t\} = -3e^{-\pi s} \cdot \frac{s}{s^2 + 1} = \frac{-3se^{-\pi s}}{s^2 + 1}$

Exercise: If $f(t) = \begin{cases} -t^2 + 3t - 2, & 1 \leq t < 2 \\ 0, & t > 2 \end{cases}$, then find $L\{f(t)\}$

THE INVERSE LAPLACE TRANSFORM

If $F(s) = L\{f(t)\}, s > s_0$ and $t \geq 0$, then $f(t)$ is called the inverse Laplace transform of $F(s)$ and is denoted by $L^{-1}\{F(s)\} = f(t), t \geq 0 \Leftrightarrow L\{f(t)\} = F(s), s > s_0$

L^{-1} -is the inverse lapltian operator

Theorem: L^{-1} is linear $L^{-1}\{c_1 F(s) + c_2 G(s)\} = c_1 L^{-1}\{F(s)\} + c_2 L^{-1}\{G(s)\}$

Example: a) $L^{-1}\left\{\frac{1}{s-a}\right\} = e^{at}, s > a, t \geq 0$. Example: $L^{-1}\left\{\frac{1}{s-3}\right\} = e^{3t}, s > 3$

b) $L^{-1}\left\{\frac{1}{s^n}\right\} = \frac{t^{n-1}}{(n-1)!}, s > 0, t \geq 0, n = 0, 1, 2, \dots$

Example: $L^{-1}\left\{\frac{1}{s^5}\right\}$. Solution: we identify $n+1=5$ or $n=4$ and then multiply and divide by $4!$. That is,

$$\frac{1}{s^5} = \frac{4!}{4!s^5} \Rightarrow L^{-1}\left\{\frac{1}{s^5}\right\} = L^{-1}\left\{\frac{4!}{4!s^5}\right\} = \frac{1}{4!} L^{-1}\left\{\frac{4!}{s^5}\right\} = \frac{1}{4!} t^4 = \frac{t^4}{24}$$

$$L^{-1}\left\{\frac{1}{s^2 + a^2}\right\} = L^{-1}\left\{\frac{a}{s^2 + a^2} \cdot \frac{1}{a}\right\} = \frac{1}{a} L^{-1}\left\{\frac{a}{s^2 + a^2}\right\} = \frac{\sin at}{a}, s > 0, t \geq 0$$

$$\text{In particular } L^{-1}\left\{\frac{1}{s^2 + 9}\right\} = L^{-1}\left\{\frac{3}{s^2 + 3^2} \cdot \frac{1}{3}\right\} = \frac{1}{3} L^{-1}\left\{\frac{3}{s^2 + 3^2}\right\} = \frac{\sin 3t}{3}, s > 0, t \geq 0$$

Example: Find a) $L^{-1}\left\{\frac{s+7}{s^2 - s - 2}\right\}$

$$\text{Solution: } \frac{s+7}{s^2 - s - 2} = \frac{s+7}{(s-2)(s+1)} = \frac{A}{s-2} + \frac{B}{s+1} = \frac{(A+B)s + A - 2B}{(s-2)(s+1)}$$

$$\Leftrightarrow s+7 = (A+B)s + A - 2B \Rightarrow A=3, B=-2$$

$$\text{Thus } L^{-1}\left\{\frac{s+7}{s^2 - s - 2}\right\} = L^{-1}\left\{\frac{3}{s-2} + \frac{-2}{s+1}\right\} = 3L^{-1}\left\{\frac{1}{s-2}\right\} - 2L^{-1}\left\{\frac{1}{s+1}\right\} = 3e^{2t} - 2e^{-t}, s > 2$$

$$\text{b) } L^{-1}\left\{\frac{s+1}{s^3 + s}\right\}$$

Solution: $\frac{s+1}{s^3+s} = \frac{s+1}{s(s^2+1)} = \frac{A}{s} + \frac{Bs+C}{s^2+1} = \frac{(A+B)s^2+Cs+A}{s(s^2+1)} \Rightarrow A=1, B=-1, C=1$

Thus $L^{-1}\left\{\frac{s+1}{s^3+s}\right\} = L^{-1}\left\{\frac{1}{s}\right\} + L^{-1}\left\{\frac{-s}{s^2+1}\right\} + L^{-1}\left\{\frac{1}{s^2+1}\right\} = 1 - \cos t + \sin t, s > 0, t \geq 0$

Theorem: Let $L^{-1}\{F(s)\} = f(t)$, then

a) $L^{-1}\{F(s-a)\} = e^{at} f(t)$ b) $L^{-1}\left\{\frac{d^n}{ds^n} F(s)\right\} = (-1)^n t^n L^{-1}\{F(s)\} = (-1)^n t^n f(t)$

c) $L^{-1}\{e^{-as} F(s)\} = f(t-a)u(t-a)$

Example: Find a) $L^{-1}\left\{\frac{s+2}{s^2+6s+10}\right\}$

Solution: Observe that $s^2+6s+10$ is not factorizable

Hence by completing the square $s^2+6s+10 = (s+3)^2+1$

Thus $\frac{s+2}{s^2+6s+10} = \frac{s+2}{(s+3)^2+1} = \frac{s+3-1}{(s+3)^2+1} = \frac{s+3}{(s+3)^2+1} - \frac{1}{(s+3)^2+1}$

$\Rightarrow L^{-1}\left\{\frac{s+2}{s^2+6s+10}\right\} = L^{-1}\left\{\frac{s+3}{(s+3)^2+1}\right\} = L^{-1}\left\{\frac{s+3-1}{(s+3)^2+1}\right\} = L^{-1}\left\{\frac{s+3}{(s+3)^2+1}\right\} - L^{-1}\left\{\frac{1}{(s+3)^2+1}\right\}$
 $= e^{-3t} \cos t - e^{-3t} \sin t$

b) $L^{-1}\left\{\frac{1}{(s-1)^2}\right\}$

Solution: $L^{-1}\left\{\frac{1}{(s-1)^2}\right\} = (-1)^1 t^1 L^{-1}\left\{\frac{-1}{s-1}\right\} = te^t$

c) Find $L^{-1}\left\{\frac{e^{-3s}}{(s-1)^4}\right\}$. Solution: $F(s) = \frac{1}{(s-1)^4}$, and

$L^{-1}\left\{\frac{1}{(s-1)^4}\right\} = e^t L^{-1}\left\{\frac{1}{s^4}\right\} = e^t \cdot \frac{1}{3!} L^{-1}\left\{\frac{3!}{s^4}\right\} = e^t \cdot \frac{1}{3!} t^3 = f(t)$. Hence, using part (III) of above theorem,

$L^{-1}\left\{\frac{e^{-3s}}{(s-1)^4}\right\} = f(t-3)u(t-3) = \frac{e^{t-3}(t-3)^3}{6} u(t-3) = \begin{cases} \frac{e^{t-3}(t-3)^3}{6}, & t \geq 3 \\ 0, & 0 \leq t < 3 \end{cases}$

Exercise: Find a) $L^{-1}\left\{\frac{1}{s^2(s+1)}\right\}$ b) $L^{-1}\left\{\frac{1}{(s-2)^3}\right\}$ c) $L^{-1}\left\{\frac{1}{(s^2+9)^2}\right\}$ d) $L^{-1}\left\{\frac{2e^{-s}}{s^2+4}\right\}$

Theorem: 1. If $f(t+T) = f(t)$, that is, f is periodic function of period T , then

$$L\{f(t)\} = \frac{1}{1 - e^{-sT}} \int_0^T e^{-st} f(t) dt$$

$$2. \text{ If } L\{f(t)\} = F(s), \text{ then } L\left\{\frac{f(t)}{t}\right\} = \int_0^\infty F(s) ds$$

Example: 1. $f(t) = \begin{cases} 1, 0 \leq t < \pi \\ -1, \pi \leq t < 2\pi \end{cases}$ and $f(t + 2\pi) = f(t)$, f is a periodic with period 2π .

Solution: $L[f(t)] = \frac{1}{1 - e^{-2\pi s}} \int_0^{2\pi} e^{-st} f(t) dt$ and

$$\begin{aligned} \int_0^{2\pi} e^{-st} f(t) dt &= \int_0^\pi e^{-st} dt - \int_\pi^{2\pi} e^{-st} dt = -\frac{1}{s} e^{-st} \Big|_0^\pi + \frac{1}{s} e^{-st} \Big|_\pi^{2\pi} \\ &= -\frac{1}{s} (e^{-s\pi} - 1) + \frac{1}{s} (e^{-2\pi s} - e^{-s\pi}) = \frac{1}{s} + \frac{1}{s} (e^{-2\pi s} - 2e^{-s\pi}) \end{aligned}$$

2. Find $L\left\{\frac{\sin t}{t}\right\}$. Solution: $L\{\sin t\} = \frac{1}{s^2 + 1} = F(s)$, we have

$$L\left\{\frac{\sin t}{t}\right\} = \int_s^\infty F(s) ds = \int_s^\infty \frac{ds}{s^2 + 1} = \lim_{T \rightarrow \infty} \int_s^T \frac{ds}{s^2 + 1}$$

$$\begin{aligned} &= \lim_{T \rightarrow \infty} \left[\tan^{-1}(s) \Big|_s^T \right] = \lim_{T \rightarrow \infty} (\tan^{-1}(T) - \tan^{-1}(s)) \\ &= \frac{\pi}{2} - \tan^{-1}(s) \end{aligned}$$

Exercise: Find $L\left\{\frac{1 - e^t}{t}\right\}$

CONVOLUTION MAPPING

Definition: Let f, g be piecewise continuous. Then the convolution of f and g is

$$(f * g)(t) = \int_0^t f(t-u)g(u) du$$

Properties: 1) $f * g = g * f$

$$2) (f * g) * h = f * (g * h)$$

$$3) f * (g + h) = f * g + f * h$$

$$4) f * (kg) = k(f * g)$$

Theorem (Convolution theorem) If $L\{f(t)\} = F(s)$ and $L\{g(t)\} = G(s)$

$$1) L\{f * g\} = Lf \cdot Lg = F(s)G(s) \quad 2) L^{-1}\{F(s)G(s)\} = (f * g)(t)$$

Example: Find a) $L^{-1}\left\{\frac{1}{s(s^2 + 9)}\right\}$ b) $L^{-1}\left\{\frac{1}{s^2(s+1)}\right\}$ c) $L^{-1}\left\{\frac{s}{(s^2 + 9)^2}\right\}$ d) $L^{-1}\left\{\frac{1}{(s^2 + 1)^2}\right\}$

Solution: Let $F(s) = \frac{1}{s}$ and $G(s) = \frac{1}{s^2 + 9} \Rightarrow L^{-1}\{F(s)\} = 1$ and $L^{-1}\{g(s)\} = \frac{\sin 3t}{3}$

$$\text{Thus } L^{-1}\left\{\frac{1}{s(s^2 + 9)}\right\} = L^{-1}\{F(s)G(s)\} = (f * g)(t) = \int_0^t f(t-u)g(u)du = \int_0^t \frac{\sin 3u}{3} du = \frac{1 - \cos 3t}{9}$$

Similarly b), c) and d) as exercise

LAPLACE TRANSFORMS OF DERIVATIVES AND INTEGRALS

Suppose $f(t)$ and its derivatives up to n -order are continuous on $[0, \infty)$ with f and each derivative having Laplace transformation. If $L\{f(t)\} = F(s)$, then

$$1) L\left\{\frac{d^n}{ds^n} f(t)\right\} = L\{f^{(n)}\} = s^n F(s) - s^{n-1} f(0) - s^{n-2} f'(0) - \dots + s^{n-k-1} f^{(k)}(0) + f^{(n-1)}(0)$$

$$\text{Here } f^{(n)}(0) = \left. \frac{d^n}{dt^n} f(t) \right|_{t=0}$$

$$\text{In particular, } L\left\{\frac{d}{ds} f(t)\right\} = L\{\dot{f}(t)\} = sL\{f(t)\} - f(0) = sF(s) - f(0)$$

$$L\left\{\frac{d^2}{ds^2} f(t)\right\} = s^2 F(s) - s f(0) - f'(0), \text{ and } L\left\{\frac{d^3}{ds^3} f(t)\right\} = s^3 F(s) - s^2 f(0) - s f'(0) - f^{(2)}(0) \text{ etc}$$

$$2) L\left\{\int_0^t f(x)dx\right\} = \frac{F(s)}{s}, \text{ and conversely } L^{-1}\left\{\frac{F(s)}{s}\right\} = \int_0^t f(x)dx$$

Example: Find a) $L\left\{\int_0^t x \cos ax dx\right\}$ b) $L^{-1}\left\{\frac{1}{s(s^2 + 9)}\right\}$ c) $L^{-1}\left\{\frac{s+1}{s^3 + s}\right\}$

$$\text{Solution: a) } L\{t \cos at\} = \frac{s^2 - a^2}{(s^2 + a^2)^2} = F(s), s > 0$$

$$\Rightarrow L\left\{\int_0^t x \cos ax dx\right\} = \frac{F(s)}{s} = \frac{s^2 - a^2}{s(s^2 + a^2)^2}, s > 0$$

$$\text{b) } \frac{1}{s(s^2 + 9)} = \frac{F(s)}{s}, F(s) = \frac{1}{s^2 + 9} \text{ and } L^{-1}\{F(s)\} = \frac{\sin 3t}{3}$$

$$\text{Thus } L^{-1}\left\{\frac{1}{s(s^2 + 9)}\right\} = L^{-1}\left\{\frac{F(s)}{s}\right\} = \int_0^t \frac{\sin 3x}{3} dx = \frac{1 - \cos 3t}{9} \quad \text{c) exercise.}$$

Solution of Initial value problem (IVP) by Laplace Transform (Application of LT)

We now apply the notion of Laplace transform of derivative and inverse Laplace transform to solve initial value problem.

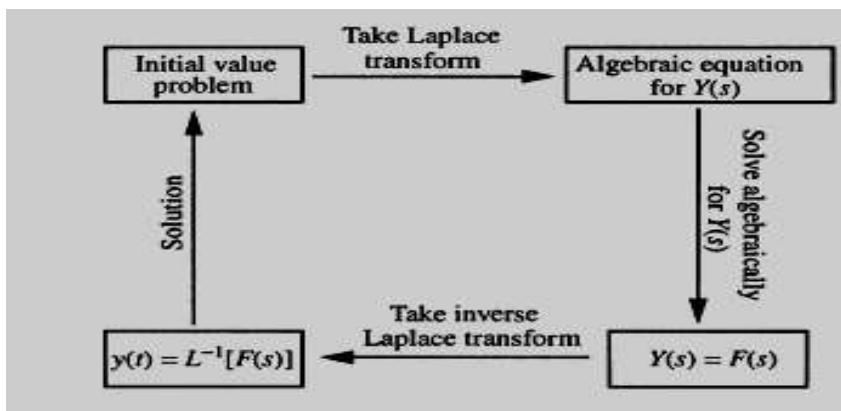
Consider the IVP $\ddot{y} + a\dot{y} + by = f(t)$, with $y(0) = y_0, y'(0) = y_1$, where a, b, y_0 and y_1 are all constants and $f(t)$ is a function of t having a Laplace transform.

Procedures (method of solution to DE by LT)

Step 1: Take the Laplace transform of the given DE, and substitute in the given conditions

Step 2: Solve the resulting equation algebraically for $Y(s)$.

Step 3: Take the inverse Laplace transform of the resulting equation to determine the solution $y(t)$ of the given IVP.



A schematic representation of the Laplace transforms method for solving IVP.

Example: Solve the IVP a) $\dot{y} + 3y = t, y(0) = 2$

Solution: Step 1: $L\{\dot{y} + 3y\} = L\{t\} \Rightarrow sL\{y(t)\} - y(0) + 3L\{y(t)\} = \frac{1}{s^2}$

$$\text{Thus } sL\{y(t)\} - 2 + 3L\{y(t)\} = \frac{1}{s^2}$$

$$\text{Step 2: } L\{y(t)\} = \frac{2s^2 + 1}{s^2(s + 3)}$$

Step 3: $y(t) = L^{-1} \frac{2s^2 + 1}{s^2(s+3)} = \frac{-5e^{-3t}}{9} + \frac{5t}{3} + \frac{5}{9}$ is the particular solution

b) $\ddot{y} - 3\dot{y} + 2y = 4t, y(0) = 1, y'(0) = 3$

Solution: step1: $L\{\ddot{y} - 3\dot{y} + 2y\} = L\{4t\}$

$$\Rightarrow s^2 L\{y(t)\} - sy(0) - y'(0) - 3[sL\{y(t)\} - y(0)] + 2L\{y(t)\} = \frac{4}{s^2}$$

$$\Rightarrow s^2 L\{y(t)\} - s - 3 - 3[sL\{y(t)\} - 1] + 2L\{y(t)\} = \frac{4}{s^2}$$

Step 2: $L\{y(t)\} = \frac{s^3 - 6s + 4}{s^2(s^2 - 3s + 2)}$

Step 3: $y(t) = L^{-1} \left\{ \frac{s^3 - 6s + 4}{s^2(s^2 - 3s + 2)} \right\}$ and

$$\begin{aligned} \frac{s^3 - 6s + 4}{s^2(s^2 - 3s + 2)} &= \frac{(s^2 + 2s - 2)(s - 2)}{s^2(s - 1)(s - 2)} = \frac{s^2 + 2s - 2}{s^2(s - 1)} \\ &= \frac{s^2}{s^2(s - 1)} + \frac{2s - 2}{s^2(s - 1)} = \frac{1}{s - 1} + \frac{2}{s^2} \end{aligned}$$

$$\Rightarrow y(t) = L^{-1} \left\{ \frac{s^3 - 6s + 4}{s^2(s^2 - 3s + 2)} \right\} = L^{-1} \left\{ \frac{1}{s - 1} + \frac{1}{s^2} \right\} = L^{-1} \left\{ \frac{1}{s - 1} \right\} + L^{-1} \left\{ \frac{2}{s^2} \right\} = e^t + 2t$$

Exercise: Using Laplace transforms solve the IVPs

a) $\ddot{y} - 5\dot{y} + 4y = e^{2t}, y(0) = 1, y'(0) = 0$ b) $\ddot{y} - 6\dot{y} + 5y = 0, y(1) = 0, y'(1) = 2$

c) $\ddot{y} + 2\dot{y} + 5y = e^{-t} \sin t, y(0) = 0, y'(0) = 1$ d) $\dot{y} - y = t^2, y(0) = 1$

e) $\dot{y} - \int_0^t y(x) dx = t^2, y(0) = 1$

f) $\ddot{y} - 6\dot{y} + 5\int_0^t y(t) dt = 0, y(1) = 0, y'(1) = 2$

g) $\dot{y} - y = \begin{cases} t^2, & t > 2 \\ -t, & t \leq 2 \end{cases}, y(0) = 1$

h) d) $\dot{y} - y = U(t - 3), y(0) = 1$

i) d) $\dot{y} - y = \begin{cases} \cos(\pi - t), & t > 0 \\ 0, & t \leq 0 \end{cases}, y(0) = 1$ j) $\dot{y} + 3y + 2\int_0^t y(t) dt = 2[u(t - 1) - U(t - 2)], y(0) = 1$